Emily R. Davenport

Assistant Professor – Pennsylvania State University
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Education

2009 - 2014 Ph.D. in Human Genetics
Certificate in University Teaching
University of Chicago; Chicago, IL
Advisor: Yoav Gilad (supported by NIH training grant)

2003 - 2007 B.S. in Bacteriology (with comprehensive honors)
University of Wisconsin – Madison; Madison, WI

2005 International Study Program, selected participant
National University of Ireland, Galway; Galway, Ireland

Academic, Research, and Industry Experience

2019 - **Assistant Professor**. Department of Biology, Huck Institutes of the Life Sciences, Institute for Computational and Data Sciences; The Pennsylvania State University; University Park, PA.

2018 - 2019 **Visiting Post-doctoral Fellow**. Supervisor: Ruth Ley (supported by NIH NRSA)
Department of Microbiome Science, Max Planck Institute for Developmental Biology; Tübingen, Germany.

2014 - 2019 **Post-doctoral Fellow**. Advisor: Andrew Clark (supported by NIH NRSA)
Department of Molecular Biology and Genetics, Cornell University; Ithaca, NY.

2008 - 2009 **Sequence Capture Technician**. Supervisor: Mindy Bennett.
Roche NimbleGen; Madison, WI.

2007 - 2008 **Forensic Scientist – DNA Analyst**. Supervisor: Sherry Culhane.
DNA Unit, Department of Justice - Wisconsin State Crime Lab; Madison, WI.

2004 - 2007 **Undergraduate Research Assistant/Independent Research**. Advisor: Steven Barclay
Department of Bacteriology, University of Wisconsin – Madison; Madison, WI.

2006 **International Research Experience for Students (IRES) in Microbiology Summer Research Program Participant**. Advisor: Sukathida Ubol
Department of Microbiology, Mahidol University; Bangkok, Thailand.

Publications – peer reviewed

* denotes equal contribution

Underlined denotes mentee author

Submitted

41. Suzuki TA, Moeller AH, Davenport ER, and Ley RE. *Response to Good (2026): Mechanisms and the gradient of population-level codiversification.*

40. Ryu EP, Keller CA, Nichols RG, Tran HN, Brocious PR, Haris LR, Koltun WA, Yochum GS, and **Davenport ER**. *Integrative host transcriptomic and mucosal microbiome profiling reveals region-specific host-microbiome associations across the human intestine*. bioRxiv: https://doi.org/10.64898/2026.05.13.725025

39. Tian Y, Koo I, Anderson BD, Vairagkar A, Anitha M, Santos A, Tian S, Antonian EA, Paudel D, Oldro KJ, Patel AD, Mullins CE, Sallam L, Koshkin S, Yennawar HP, Fecko JA, Cutler T, Gray JL, Ning G, Hyun Cho S, Zhao J, Swencki-Underwood B, Imchen M, Nichols RG, Hao F, Weiss AS, **Davenport ER**, Zackular JP, Bordenstein SR, Walker DI, Yennawar NH, Alderete TL, Weinert EE, Peters JM, Bisanz JE, and Patterson AD. *PFAS drive gut dysbiosis and antibiotic resistance.*

38. Nichols RG, Boyle N, Hao F, Anitha M, Tian Y, Peters JM, **Davenport ER**, and Patterson AD. *Short-term chlorpyrifos exposure alters gut microbiome and metabolome in mice*.

2026

37. Tran HN, Kirven KJ, and **Davenport ER**. *SCiMS: Sex calling in metagenomic sequences*. bioRxiv: https://www.biorxiv.org/content/10.64898/2026.02.17.705110v1 [accepted Microbiome]

36. Ramachandran SL*, Pasupuleti N*, Abdil RJ†, Adikari G†, Ahlawat B†, Blekhman R†, Burgo V†, **Davenport ER**†, Dwivedi A†, Gupta S†, Ijinu TP†, Jamir T†, Karu R†, Kaushik A†, Komarabthini JM†, Laithangpuii†, Lalzarliana J†, Norbu T†, Pautu T†, Pawar K†, Raman AS†, Ramaswamy R†, Ranasinghe R†, Rashmi†, Sasidharan SP†, Saheesh T†, Shaji S†, Shapiro JW†, Singh E†, Singson V†, Sundararajan A†, Tennekoon KH†, Tetso D†, Urban Aragon JA†, Welikala AHJ†, Yhome RR†, Rai N, and Raghavan M. *Distinct trajectories of urbanization shape the human gut microbiome across South Asia*. Gut Microbes. 2026;18(1). DOI: https://doi.org/10.1080/19490976.2026.2700042

    † These authors listed alphabetically

35. Tikhonova P, Mekonnen Z, Belanger KD, Ay A, **Davenport ER**, and Taye B. *Microbiome responses to anthelmintic treatment depend on pre-treatment helminth infection status in young Ethiopian children*. Communications Medicine. 2026. DOI: https://doi.org/10.1038/s43856-026-01625-3

2025

34. Kelliher JM, Mirzayi C, Bordenstein SR, Oliver A, Kellogg CA, Hatcher EL, Berg M, Baldrian P, Aljumaah M, Maria Luz Miller C, Mungall C, Noval V, Palucki A, Smith E, Tabassum N, Bonito G, Brister JR, Chain PSG, Chen M, Degregori S, Pabo Dundore-Arias J, Emerson JB, Moreira C. Fernandes V, Flores R, STREAMS Consortium, and Eloe-Fadrosh EA [**Davenport ER** part of STREAMS Consortium]. S*TREAMS guidelines: standards for technical reporting in environmental and host-associated microbiome studies.* Nature Microbiology. 2025;10(3059-3068). DOI: https://doi.org/10.1038/s41564-025-02186-2

33. Van Syoc EP, Gomez A, **Davenport ER**, and Bordenstein SR. *Gut fungal profiles reveal phylosymbiosis and codiversification across humans and nonhuman primates.* PLoS Biology 23(9):e3003390. DOI: https://doi.org/10.1371/journal.pbio.3003390

32. Van Syoc EP, **Davenport ER**, and Bordenstein SR. *Gut fungi are associated with human genetic variation and disease risk*. PLoS Biology 23(9):e3003339. DOI: https://doi.org/10.1371/journal.pbio.3003339

31. Cappas V, Roy R, **Davenport ER**, and Sykes D. *The microbiome and volatile organic compounds reflect the state of decomposition in an indoor environment.* Science & Justice. 2025;65(5):101302. DOI: https://doi.org/10.1016/j.scijus.2025.101302

30. Chilkina M, Tran H, Uribe G, McAninch S, McAninch C, Seideneck A, He B, Khanipov K, Golovko G, Powell DW, **Davenport ER,** Pinchuk IV. *MyD88-mediated signaling in intestinal fibroblasts regulates macrophage antimicrobial defense and prevents dysbiosis in the gut*. Cell Reports. 2025;44(5):115553. DOI: https://doi.org/10.1016/j.celrep.2025.115553

29. Litichevskiy L, Considine M, Gill J, Shandar V, Cox TO, Descamps HC, Wright KM, Amses KR, Dohnalova L, Liou MJ, Tetlak M, Galindo-Fiallos MR, Wong AC, Lundgren P, Kim J, Uhr GT, Rahman RJ, Mason S, Merenstein C, Bushman FD, Raj A, Harding F, Chen Z, Prateek GV, Mullis M, Deighan AG, Robinson L, Tanes C, Bittinger K, Chakraborty M, Bhatt AS, Li H, Barnett I, **Davenport ER**, Broman KW, Cohen RL, Botstein D, Freund A, Di Francesco A, Churchill GA, Li M, and Thaiss CA. *Gut metagenomes reveal interactions between dietary restriction, aging, and the microbiome in genetically diverse mice*. Nature Microbiology. 2025;10,1240-1257. DOI: https://doi.org/10.1038/s41564-025-01963-3

28. Liu S, Rodriguez J, Munteanu V, Ronkowski C, Kumar Sharma N, Alser M, Andreace F, Blekhman R, Błaszczyk D, Chikhi R, Crandall KA, Libera KD, Francis D, Frolova A, Gancz AS, Huntley NE, Jaiswal P, Kosciolek T, Łabaj P, Łabaj W, Luan T, Mason C, Moustafa AM, Muralidharan HS, Mutlu O, Ghiasi NM, Rahnavard A, Sun F, Tian S, Tierney BT, Van Syoc E, Vicedomini R, Zackular JP, Zelikovsky A, Zielińska, Ganda E, **Davenport ER**, Pop M, Koslicki D, Mangul S. *Analysis of metagenomic data*. Nature Reviews Methods Primers. 2025;5(5). DOI: https://doi.org/10.1038/s43586-024-00376-6

2024

27. Nichols RG and **Davenport ER**. *Clade-specific long-read sequencing increases the accuracy and specificity of the gyrB phylogenetic marker gene*. mSystems. 2024;10:e01480-24. DOI: https://doi.org/10.1128/msystems.01480-24

26. Ryu EP, Gautam Y, Proctor DM, Bhandari D, Tandukar S, Gupta M, Prasad Gautam G, Relman DA, Shibl AA, Bahadur Sherchand J, Jha AR, and **Davenport ER**. *Nepali oral microbiomes reflect a gradient of lifestyles from traditional to industrialized.* Microbiome. 2024;12,228. DOI: https://doi.org/10.1186/s40168-024-01941-7

25. Nichols RG, Rimal B, Hao F, Peters JM, **Davenport ER**, and Patterson AD. *Chlorpyrifos modulates the mouse gut microbiota and metabolic activity*. Environment International. 2024;192:109022. DOI: https://doi.org/10.1016/j.envint.2024.109022

24. Taye B, Mekonnen Z, Belanger KD, and **Davenport ER**. *Gut-microbiome profiles among Soil-Transmitted Helminths (STHs) infected Ethiopian children enrolled in the school-based mass deworming program.* PLoS Neglected Tropical Diseases 18(10), e0012485. DOI: https://doi.org/10.1371/journal.pntd.0012485

2023

23. Gancz AS, Farrer AG, Nixon M, Arriola L, Adler C, **Davenport ER**, Gully N, Cooper A, Britton K, Dobney K, Silverman J, Weyrich L. *Ancient dental calculus reveals oral microbiome shifts associated with lifestyle and disease in Great Britain.* Nature Microbiology. 2023;8,2315-2325. DOI: https://doi.org/10.1038/s41564-023-01527-3

22. Kelliher JM, Rudolph M, Vangay P, Abbas A, Borton MA, Santistevan C, **Davenport ER,** Davenport DW, Erazo NG, Herman C, Karstens L, Kocurek B, Lutz HL, Myers KS, Ockert I, Rodriguez FE, Saunders JK, Smith ML, Vogtmann E, Windsor A, Wood-Charlson EM, Woodley L, Eloe-Fadrosh EA. *Cohort-based learning for microbiome community*. Nature Microbiology. 2023;8, 751-753. DOI: https://doi.org/10.1038/s41564-023-01361-7

2022

21. Ryu EP and **Davenport ER**. *Host genetic determinants of the microbiome across animals: from C. elegans to cattle.* Annual Reviews of Animal Biosciences. 2022 10:1, 203-226. DOI: https://doi.org/10.1146/annurev-animal-020420-032054

2021

20. Jeganathan NA, **Davenport ER**, Yochum GS, Koltun WA. *The microbiome in diverticulitis*. Current Opinion in Physiology. 2021;22. DOI: https://doi.org/10.1016/j.cophys.2021.06.006

19. Amato KR, Arrieta M, Azad M, Bailey MT, Broussard JL, Bruggeling C, Claud EC, Costello EK, **Davenport ER**, Dutilh BE, Swain Ewald HA, Ewald P, Hanlon EC, Julion WA, Keshavarzian A, Maurice CF, Miller GE, Preidis GA, Ségurel L, Singer BH, Subramanian S, Zhao L, Kuzawa CW.

The human microbiome and health inequities. PNAS. 2021;118(25). DOI: https://doi.org/10.1073/pnas.2017947118

18. Marderstein AR, **Davenport ER**, Kulm S, Van Hout C, Elemento O, Clark AG. *Leveraging phenotypic variability to identify genetic interactions in human phenotypes.* American Journal of Human Genetics. 2021;108(1). DOI: https://doi.org/10.1016/j.ajhg.2020.11.016

2020

17. Nichols RG and **Davenport ER.** *The relationship between the gut microbiome and host gene expression: a review*. Human Genetics. 140(5), 747-760. DOI: 10.1007/s00439-020-02237-0

2019

16. Gong X, **Davenport ER**, Wang D, Clark AG. *Lack of spatial and temporal genetic structure of Anguilla japonica populations.* Conservation Genetics. 2019;20:467. DOI: https://doi.org/10.1007/s10592-019-01146-8

15. Kachoo P*, Eraso JM*, Beres SB, Olsen RJ, Zhu L, Nasser W, Bernard PE, Cantu CC., Ojeda Saavedra M, José Arredondo M, Strope B, Do H, Kumaraswami M, Vuopio J, Gröndahl-Yli-Hannuksela K, Kristinsson KG, Gottfredsson M, Pesonen M, Pensar J, **Davenport ER**, Clark AG, Corander J, Caugant DA, Gaini S, Magnusses MD, Porter AR, DeLeo FR, and Musser JM. *Integrated analysis of population genomics, transcriptomics and virulence provides novel insights into* Streptococcus pyogenes *pathogenesis.* Nature Genetics. 2019;51 548-559. DOI: https://doi.org/10.1038/s41588-018-0343-1

2018

14. Jha AR, **Davenport ER**, Gautam Y, Bhandari D, Tandukar S, Ng K, Holmes S, Prasad Gautam G, Bahadur Sherchand J, Bustamante CD, and Sonnenburg JL. *Gut microbiome transition across a life gradient in Himalaya.* PLoS Biology. 2018;16(11):e2005396 (epub 2018 Nov 15). DOI: https://doi.org/10.1371/journal.pbio.2005396

2017

13. **Davenport ER***, Sanders JG*, Song SJ, Amato KR, Clark AG, and Knight R. *The human microbiome in evolution*. BMC Biology. 2017. 15:127. DOI: https://doi.org/10.1186/s12915-017-0454-7

12. Goodrich JK, **Davenport ER,** Clark AG, and Ley RE. *The relationship between the human genome and microbiome comes into view*. Annual Review of Genetics. 2017. 51(1). DOI: https://doi.org/10.1146/annurev-genet-110711-155532

11. Igartua C, **Davenport ER**, Gilad Y, Nicolae DL, Pinto J, and Ober C. *Host genetic variation in mucosal immunity pathways influences the upper airway microbiome*. Microbiome. 2017 Feb 1;5:16. DOI: https://doi.org/10.1186/s40168-016-0227-5

2016

10. **Davenport ER**, Goodrich JK, Bell JT, Spector TD, Ley RE, Clark AG. *ABO antigen and secretor statuses are not associated with gut microbiota composition in 1,500 twins*. BMC Genomics. 2016 Nov 21;17:941. DOI: https://doi.org/10.1186/s12864-016-3290-1

9. Beaumont M, Goodrich JK, Jackson MA, Yet I, **Davenport ER**, Vieira-Silva S, Debelius J, Pallister T, Mangino M, Raes J, Knight R, Clark AG, Ley RE, Spector TD, and Bell JT. *Heritable components of the human fecal microbiome are associated with visceral fat*. Genome Biology. 2016 Sep 26;17:189. DOI: https://doi.org/10.1186/s13059-016-1052-7

8. Goodrich JK, **Davenport ER**, Beaumont M, Jackson MA, Knight R, Spector TD, Bell JT, Clark AG, and Ley RE. *Genetic determinants of the gut microbiome in UK twins*. Cell Host and Microbe. 2016: 19(5), 731-743. DOI: https://doi.org/10.1016/j.chom.2016.04.017

7. Goodrich JK*, **Davenport ER***, Waters JL*, Clark AG, and Ley RE. *Cross-species comparisons of host genetic associations with the microbiome*. Science. 2016: 352(6285), 532-535. DOI: https://doi.org/10.1126/science.aad9379

6. **Davenport ER**. *Elucidating the role of the host genome in shaping microbiome composition*. Gut Microbes. 2016: 7(2), 178-184. DOI: https://doi.org/10.1080/19490976.2016.1155022

5. Blischak JD, **Davenport ER**, and Wilson G. *A quick introduction to version control with Git and GitHub*. PLoS Computational Biology. 2016;12(1):e1004668. DOI: https://doi.org/10.1371/journal.pcbi.1004668

2015

4. **Davenport ER,** Cusanovich DA, Michelini K, Barreiro LB, Ober C, and Gilad Y. *Genome-wide association studies of the human gut microbiota*. PLoS One. 2015;10(11):e0140301 (epub 2015 Nov 3). DOI: https://doi.org/10.1371/journal.pone.0140301

   -----> *An Editor's Pick for the PLoS Microbiology special collection:*
   http://collections.plos.org/microbiology-picks

   -----> *An Editor's Pick for the PLoS Experimental Biology special collection:*
   http://collections.plos.org/experimental-biology

2014

3. Zhou X, Cain CE, Myrthil M, Lewellen N, Michelini K, **Davenport ER**, Stephens M, Pritchard JK, and Gilad Y. *Epigenetic modifications are associated with inter-species gene expression variation in primates*. Genome Biology. 2014 Dec 3;15(12):547. DOI: https://doi.org/10.1186/s13059-014-0547-3

2. **Davenport ER**, Mizrahi-Man O, Michelini K, Barreiro LB, Ober C, and Gilad Y. *Seasonal variation in human gut microbiome composition*. PLoS One. 2014;9(3):e90731 (epub 2014 Mar 11). DOI: https://doi.org/10.1371/journal.pone.0090731

2013

1. Mizrahi-Man O, **Davenport ER**, and Gilad Y. *Taxonomic classification of bacterial 16S rRNA genes using short sequencing reads: Evaluation of effective study designs*. PLoS One. 2013;8(1):e53608 (epub 2013 Jan 7). DOI: https://doi.org/10.1371/journal.pone.0053608

Publications – non-peer reviewed

4. Chen YY, Evavold CL, Mann M, **Davenport ER**, McFall-Ngai M, Bienko M, Ueda HR, Tian L, Tjahjono N, Anikeeva P, Gogo Liu J, Deans TL, and Shen X. *What tool or method do you wish existed?* Cell. 2024;187(17). DOI: https://doi.org/10.1016/j.cell.2024.07.043

3. Ryu EP and **Davenport ER**. *Microbiome: Taming the beasts inside*. eLife 2021;10:e67634. DOI: https://doi.org/10.7554/eLife.67634

2. **Davenport ER**. *Genetic variation shapes murine gut microbiota via immunity*. Trends in Immunology. 2020;41(1). DOI: https://doi.org/10.1016/j.it.2019.11.009

1. **Davenport ER**. *Tooth be told, genetics influence oral microbiome*. Cell Host & Microbe. 2017;22(3). DOI: https://doi.org/10.1016/j.chom.2017.08.018

Presentations

Invited Plenary Platform Presentations

2020 **Davenport, ER**. *Role of host genetics in shaping the gut microbiota*. Gut Microbiome for Health World Summit. Madrid, Spain [note: presented remotely due to COVID-19]

Invited Platform Presentations

2027 **Davenport, ER**. *What shapes and shatters host-microbiome phylosymbiosis*. Society for Integrative and Comparative Biology (SICB); Symposium: Feedback between host phenotypes and the microbiome: Physiological mechanisms and behavioral outcomes. Pittsburgh, PA

2026 **Davenport, ER**. *What shapes and shatters host-microbiome phylosymbiosis*. International Geobiology Course Symposium. Pennsylvania State University, University Park, PA

2026 **Davenport, ER**. *Host-mycobiome crosstalk reveals shared and site-specific patterning across the intestinal mucosa*. Inflammatory Bowel Disease (IBD) Research Symposium. Pennsylvania State University College of Medicine. Hershey, PA

2025 **Davenport, ER**. *Gaining insights into host-microbiome physiology via multi-omics integration*. Multi-Omics at Scale: Leveraging Penn State's Expertise for Future Interdisciplinary Opportunities Workshop at the Institute of Computational and Data Science (ICDS) Symposium. Pennsylvania State University, University Park, PA

2025 **Davenport, ER**. *Using the microbiome as a proxy for the environment in genome-wide association studies*. NYU Abu Dhabi Center in Genomics & Systems Biology Symposium. Abu Dhabi, UAE.

2024 **Davenport, ER**. *Associations between host gut epithelial gene expression and the mucosal microbiome along the lower GI tract*. Inflammatory Bowel Disease (IBD) Research Symposium. Pennsylvania State University College of Medicine. Hershey, PA

2023 **Davenport, ER.** Career Development in Quantitative Genetics and Genomics Mentorship Session. Quantitative Genetics and Genomics Gordon Research Seminar. Ventura, CA

2023 **Davenport, ER**. *Associations between microbes and host gene expression reveal physiological insights into host-microbiome interactions*. Quantitative Genetics and Genomics Gordon Research Conference. Ventura, CA

2022 **Davenport, ER**. *Using human genetics to understand the physiological basis of host microbiome interactions*. Inflammatory Bowel Disease (IBD) Research Symposium. Pennsylvania State University College of Medicine. Hershey, PA

2022 **Davenport, ER**. *Using the microbiome as a proxy for the environment in genome-wide association studies*. Human Genetic Variation and Disease Gordon Research Conference (HGVD GRC). Manchester, NH [*meeting cancelled]

2020 **Davenport, ER**. *Using human genetics to understand the physiological basis of host microbiome interactions.* The Vanderbilt Institute for Infection, Immunology, and Inflammation (VI4) Annual Symposium [*virtual due to COVID-19]

2020 **Davenport, ER**. *Using human genetics to understand the physiological basis of host microbiome interactions.* Workshop: Cultivating communities: Making sense of host-

microbiome interactions through the lens of genetics. The Allied Genetics Conference (TAGC). Washington DC [*virtual due to COVID-19]

2019 **Davenport, ER**. *Causes and consequences of human gut microbiome variation*. One Health Workshop. Pennsylvania State University. State College, PA

2019 **Davenport, ER**, Spector TD, Ley RE, and Clark AG. *Simultaneously modeling host genetics and microbiome impact on immune-related traits*. Pennsylvania State University Bioinformatics and Genomics Annual Retreat. State College, PA

2018 **Davenport, ER**, Spector TD, Ley RE, and Clark AG. *Simultaneously modeling host genetics and microbiome composition reveals the heritability and proportion of variance explained due to the microbiome for immune-related traits*. Probabilistic Modeling in Genomics (ProbGen). Cold Spring Harbor Labs, NY

* Co-chair of "Cancer, the microbiome, and beyond" session

2017 **Davenport ER**. *Role of host genetics in shaping the gut microbiota*. The Human Capital and Economic Opportunity (HCEO) Working Group Conference on The Gut Microbiome in Human Biology and Health: New Opportunities for the Study of Health Disparities. Chicago, IL

2017 **Davenport ER**. *The role of host genetics in determining human gut microbiome composition*. The American Association of Physical Anthropologists Annual Meeting. Wiley Invited Podium Symposium - Humans as Holobionts: The Microbiome as a Biological System in Human Evolution. New Orleans, LA.

2016 **Davenport ER**. *The role of host genetics in determining human gut microbiome composition*. The 2016 Nordic-North American Symposium on Antimicrobial Resistance and Molecular Population Genomics in Houston, TX.

Platform Presentations

2026 **Davenport ER**. *Host-mycobiome crosstalk reveals shared and site-specific patterning across the intestinal mucosa*. 10th Conference on Beneficial Microbes in Madison, WI

2023 **Davenport ER**, Spector TD, Ley RE, and Clark AG. *Using the microbiome as a proxy for host environmental influences in genome-wide association studies*. Microverse Symposium in Jena, Germany

2022 **Davenport ER**, Spector TD, Ley RE, and Clark AG. *Using the microbiome as a proxy for host environmental influences in genome-wide association studies*. CSHL Microbiome in Cold Spring Harbor, NY.

2017 **Davenport ER**, Spector TD, Ley RE, and Clark AG. *Co-occurrence network modeling reveals disease-specific configurations of microbiome community structure across 2,500 twins*. American Society of Human Genetics Annual Meeting (ASHG) in Orlando, FL.

2017 **Davenport ER**, Spector TD, Ley RE, and Clark AG. *Modeling human gut microbiome community structure across healthy and diseased states in 2,500 twins*. Society of Molecular Biology and Evolution Annual Meeting (SMBE) in Austin, TX.

2017 **Davenport ER**, Spector TD, Ley RE, and Clark AG. *Modeling human gut microbiome community structure across healthy and diseased states in 2,500 twins*. Biology of Genomes (BoG) in Cold Spring Harbor, NY.

2012 **Davenport ER**, Mizrahi-Man O, Barreiro LB, Ober C, and Gilad Y. *Examining the roles of diet, age, and sex on the composition of the human fecal microbiome*. University of Chicago Molecular Biosciences Cluster Retreat in Galena, IL.

Poster presentations

2023 **Davenport ER**, Kirven K, and Tran H. *SCiMS: Sex calling from metagenomic samples*. American Society of Human Genetics Annual Meeting (ASHG) in Washington, DC.

2015 **Davenport ER**, Goodrich JK, Bell JT, Spector TD, Ley RE, and Clark AG. *ABO antigen and secretor status are not associated with gut microbiota composition*. American Society of Human Genetics Annual Meeting (ASHG) in Baltimore, MD.

2014 **Davenport ER**, Mizrahi-Man O, Michelini K, Barreiro LB, Ober C, and Gilad Y. *poopQTLs: Genome-wide associations of the human gut microbiota*. Society for Molecular Biology and Evolution Annual Meeting (SMBE) in San Juan, PR.

2013 **Davenport ER**, Mizrahi-Man O, Michelini K, Barreiro LB, Ober C, and Gilad Y. *Temporal variation in human gut microbiome composition in the Hutterites*. American Society of Human Genetics Annual Meeting (ASHG) in Boston, MA.

2013 **Davenport ER**, Mizrahi-Man O, Michelini K, Barreiro LB, Ober C, and Gilad Y. *Examining the temporal stability of the fecal microbiome in an isolated, founder population*. Cell Symposium: the Microbiome and Host Health in Lisbon, Portugal.

2012 **Davenport ER**, Mizrahi-Man O, Barreiro LB, Ober C, and Gilad Y. *Examining the roles of diet, age, and sex on the composition of the human fecal microbiome*. American Society of Human Genetics Annual Meeting (ASHG) in San Francisco, CA.

2012 **Davenport ER**, Mizrahi-Man O, Barreiro LB, Ober C, and Gilad Y. *Examining the genetic basis of interindividual variation in the human fecal microbiome*. International Human Microbiome Conference in Paris, France.

2011 Yao T, **Davenport ER**, Poroyko V, Liu D, Lemanske R, Gern J, Ober C, Jackson D, Gilad Y, Pinto J. *The nasal microbiome and development of asthma in a birth cohort*. Biology of Genomes (BoG) in Cold Spring Harbor, NY.

Invited Seminars

2025	Department of Human Genetics, University of Pittsburgh
	Department of Medicine Grand Rounds, The Pennsylvania State University Milton S. Hershey College of Medicine
	Division of Gastroenterology and Hepatology, The Pennsylvania State University Milton S. Hershey College of Medicine
2024	Society for Epidemiologic Research (SER) Webinar Series (online)
2023	Department of Biology, Brigham Young University
	Department of Biology, University of Utah
	Department of Biology, West Chester University
	Rutgers University Microbiome Program, Rutgers University
	Balance of the Microverse Cluster of Excellence, Friedrich Schiller University Jena, Germany
	MCIBS Graduate Program Retreat, Pennsylvania State University
	Center of Genetic Epidemiology, University of Southern California
2022	ProK Seminar Series, University of Pennsylvania
	Your Researcher Live, Researcher (online)
2021	MCIBS Graduate Program Retreat, Pennsylvania State University
	UGA Interdepartmental Health Disparities Working Group, University of Georgia
	Department of Biology, Trinity University
	Department of Human Genetics, University of Chicago
	Department of Cellular and Molecular Medicine, University of Arizona
	Gut Interest Group, Penn State College of Medicine
	Microbiome Seminar Series, Pennsylvania State Microbiome Center, Pennsylvania State University
2020	Department of Public Health Sciences, Penn State College of Medicine
	Department of Biology, Vanderbilt University [cancelled because of COVID-19]
	Department of Human Genetics, UCLA [cancelled because of COVID-19]
	Department of Human Genetics, University of Chicago [cancelled because of COVID-19]
2019	Penn State Microbiome Center Exchange, Penn State College of Medicine
	Department of Biology, Duke University
	Department of Nutrition, Pennsylvania State University
	Department of Biology, University of New Mexico
	Department of Biology, Marquette University

Department of Biology, Auburn University
Department of Biology, Pennsylvania State University

2018 Max Planck Institute for Developmental Biology
Department of Biology, Bucknell University
Department of Biology, Barnard College
Department of Biology, Smith College

2017 Department of Biology, Wake Forest University

2016 The Huck Institutes of the Life Sciences, Pennsylvania State University

2013 Department of Biology, Chicago State University

2013 Emory University and Yerkes National Primate Research Center

Funding

Current

2022 – 2027 **R35 NIGMS/NIH (Award #R35GM146980)** ($1,938,350)
Characterizing human-microbiome interactions via molecular and functional genomic approaches
Role: PI
Life Event Supplement awarded (2023): **3R35GM146980-02S1** ($105,819)
Equipment Supplement awarded (2024): **3R35GM146980-02S2** ($24,752)

2026 – 2029 **R01 NIDDK/NIH (Award #R01DK143098)** ($2,843,812)
Mechanisms of the inflammatory activation of mesenchymal cells in ulcerative colitis
Role: co-I

Trainee-led

2024 – 2026 **NIH T32 (Award T32GM152354)**
Eukaryotic Gene Regulation (EGR) Training Program
Appointee: Hanh Tran; Molecular, Cellular, and Integrative Biosciences (MCIBS) PhD Program
Role: Mentor

2025 **NIH F32 (Award F32DK141228-01A1**)
NIH Ruth L. Kirschstein National Research Service Award (NRSA)
The genetic, symbiotic, and evolutionary basis of the human gut mycobiome
PI: Emily Van Syoc
Role: Co-sponsor with Seth Bordenstein

2024 – 2025 **NIH TL1 (Awards TL1TR002016, UL1TR002014)**

Translational Research (TL1) Training Program
Appointee: Polina Tikhonova; Bioinformatics and Genomics (BG) PhD Program
Role: Mentor

2021 – 2023 **NIH T32 (Award T32GM102057)**
Computation, Bioinformatics, and Statistics (CBIOS) Training Program
Appointee: Nicholas Locatelli, Biology PhD Program
Role: Co-mentor; Primary Mentor: Iliana Baums

2021 – 2023 **NIH T32 (Award T32GM102057)**
Computation, Bioinformatics, and Statistics (CBIOS) Training Program
Appointee: Erica Ryu, Biology PhD Program
Role: Mentor

2020 – 2023 **USDA NIFA (Award #2020-67034-31764)** ($165,000)
Investigating the effects of chronic, dietary, and environmentally relevant doses of chlorpyrifos on the gut microbiome of mice
PI: Robert Nichols
Role: Co-sponsor. Sponsor: Andrew Patterson

Completed

2022 – 2023 **Interdisciplinary Innovation Fellowship**, Microbiome Center, Pennsylvania State University ($5000)

2023 **Microbiome Center Shared Resource Proposal**; total: $30,000
Microbiome Center, Pennsylvania State University
Unique dual indexes to drive microbiome sequencing at PSU
PIs: Jordan Bisanz, Justin Silverman, Erika Ganda, and Emily Davenport

2022 **USDA Conference Grant (Award #2022-67019-36985);** total: $50,000
Changing Microbiomes: A symposium hosted by the Microbiome Center at the Pennsylvania State University
PI: Liana Burghardt
Role: Co-investigator

2022 **DOE Conference Grant (Award #DE-SC0022552**); total: $10,000
Changing Microbiomes: A symposium hosted by the Microbiome Center at the Pennsylvania State University
PI: Estelle Couradeau
Role: Co-investigator

2016 – 2019 **F32 NIDDK/NIH (Award #F32DK109595);** total: $173,089
NIH Ruth L. Kirschstein National Research Service Award (NRSA)
Modeling human gut microbiome community structure in healthy and disease states
Role: PI

2017 Genetics Society of America (GSA) DeLill Nasser Travel Award for Professional Development in Genetics; total: $1,000

2014 University of Chicago Biological Sciences Division Travel Award; total: $1,000

2010 – 2012 NIH Genetics and Regulation Training Grant T32 appointee (University of Chicago – 2 years of stipend support and tuition)

2011 University of Chicago Digestive Diseases Research Core Center (DDRCC) Pilot and Feasibility award (written by ER Davenport to support dissertation research, submitted by Y. Gilad); total: $20,000

Patents

2026 Bordenstein, Seth; Van Syoc, Emily; **Davenport, Emily**. *Methods and compositions for preventing and treating cardiovascular disease*. PCT Application No. PCT/US26/26286. Filed May 1, 2026; U.S. National State Application No. 19/628,377, filed September 25th, 2026. Patent pending.

Teaching Experience

University Courses – Instructor of Record

2020 – Pennsylvania State University, Department of Biology, *BIOL439: Practical Bioinformatics* (~25-30 students per semester)

2020 – Pennsylvania State University, Department of Biology, *PSU16: First Year Seminar* (~20-40 students per semester)

Teaching Assistantships

2014 University of Chicago, Biological Sciences Division: *HGEN 47300: Genomics and Systems Biology* (Taught 6 lectures)

2011 University of Chicago, Biological Sciences Division: *HGEN 47000: Human Genetics I*

2011 University of Chicago, Biological Sciences Division: *HGEN 47300: Genomics and Systems Biology*

2010 University of Chicago, Biological Sciences Division: *MGCB 31400: Genetic Analysis of Model Organisms*

Guest Lectures

2020; 2022 Pennsylvania State University, Anthropology *ANTH497: Microbial Insights into Anthropology – "Microbiome studies in the Hutterites"*

2016 Cornell University, Molecular Biology and Genetics: *BIOMG 4870: Human Genomics – "Cystic Fibrosis and PKU"*

2016 Cornell University, Biological Sciences: *BIOMI 3210: Human Microbes and Health – "Microbiome studies in the Hutterites"*

2016 Pennsylvania State University, Biochemistry and Molecular Biology: *BMB 484: Functional Genomics – "Introduction to Population Genetics"*

2015 Cornell University, Molecular Biology and Genetics: *BIOMG 4870: Human Genomics – "Linkage disequilibrium mapping, or Genome-wide Association Studies (GWAS)"*

2011 University of Chicago, Biological Sciences Division: *HGEN 47000: Human Genetics I – "Human genome structure and variation"*

Workshops

2024 – **Co-organizer** – "*Microbiome Center Kick-Start Workshop*", Pennsylvania State University, Huck Institutes of the Life Sciences, Microbiome Center

2020 – **Instructor** – "*Microbiome Center Kick-Start Workshop*", Pennsylvania State University, Huck Institutes of the Life Sciences, Microbiome Center

2023 **Instructor** – "*Git and GitHub*", Pennsylvania State University, Huck Institutes of the Life Sciences, Microbiome Center

2022 **Instructor** – "*NMDC Metadata Standards Workshop*", Changing Microbiomes Symposium, Pennsylvania State University

2021 **Instructor** – "*Git and GitHub*", Huck Institutes of the Life Sciences, Microbiome Center, Pennsylvania State University

2019 **Instructor** – "*Introduction to Statistics with R*", Max Planck Institute for Developmental Biology; Tübingen, Germany

2016 **Instructor** – "*Learn about Git and GitHub*", Cornell University, CPGSA

2016 **Instructor** – *"Introduction to R",* University of Chicago, Biological Sciences Division

Software Carpentry Workshops [content I taught]

2017 **Lead Instructor**, TGen, Phoenix, AZ (June) [R and version control with Git]

2016 **Instructor**, University of Chicago, Biological Sciences Division (September) [review of shell and R, writing reproducible reports, and version control with Git]

2016 **Lead Instructor**, Cornell University, Department of Molecular Biology and Genetics (August) [version control with Git]

2015 **Instructor**, University of Chicago, Biological Sciences Division (September) [review of shell and R, writing reproducible reports, and version control with Git]

2015 **Instructor**, Pennsylvania State University (June) [shell and version control with Git]

2014 **Instructor**, University of Chicago, Biological Sciences Division (September) [version control with Git]

2014 **Lead Instructor**, University of Toronto (July) [version control with Git]

2013 **Instructor**, University of Chicago, Biological Sciences Division (September) [shell]

2013 **Instructor**, University of Chicago. (June) [shell]

Data Carpentry Workshops [content I taught]

2016 **Instructor**, Cornell University (June) [reproducible reports with Rmarkdown and R programming]

2015 **Instructor**, Cornell University (January) [automating repetitive tasks with command line shell]

Mentorship

Graduate Student Advising

Years	Name	Institution	Program
2024 –	Hanh Tran	PSU	MCIBS
	---> NIH T32 EGR Training Grant Appointee (2024 - 2026)		
	---> Awarded an NSF GRFP Honorable Mention (2025)		
2022 –	Naomi Huntley	PSU	Biology
	---> Awarded the Microbiome Center Data Analysis Leadership Fund (2022)		
	---> OHMC-Qiagen Internship selectee (2025), see: https://www.psu.edu/news/huck-institutes-life-sciences/story/two-penn-state-doctoral-students-selected-industry-internship		
	---> Awarded Honorable Mention for the Fox Degree Completion Award from the J. Jeffrey and Anne Marie Fox Graduate School (2026)		
	---> Awarded Best Poster in the Human Health Category at the One Health Microbiome Center Symposium (2026)		
2022 – 2026	Polina Tikhonova	PSU	Bioinformatics and Genomics
	---> NIH TL1 Translational Research Training Program Appointee (2024 - 2025)		
	---> Awarded the Microbiome Center Data Analysis Leadership Fund (2025)		
2020 – 2025	Erica Ryu	PSU	Biology
	---> Awarded the Dr. John Randall Shuman Troxell Memorial Scholarship in Biology (2022)		
	---> Awarded the J. Ben and Helen D. Hill Memorial Fund Award (2022)		
	---> Awarded an NSF GRFP Honorable Mention (2021)		
	---> NIH T32 CBIOS Training Grant Appointee (2021 - 2022)		
2022 – 2024	Nicholas Locatelli	PSU	Biology *co-advised with Iliana Baums
	---> NIH T32 CBIOS Training Grant Appointee (2021 - 2022)		

---> *Awarded the Dr. John Randall Shuman Troxell Memorial Scholarship in Biology (2023)*

Postdoc and Scientist Advising

Years	Name	Type	Institution	Program
2023 –	Robert Nichols	Asst. Research Professor	PSU	Biology
2026	Erica Ryu	Postdoc	PSU	Biology
2024 – 2025	Emily Van Syoc	Postdoc	PSU	Biology *co-advised with Seth Bordenstein
2020 – 2023	Robert Nichols	Postdoc	PSU	Biology
	---> *Awarded a USDA NIFA Postdoctoral Fellowship (2020)*			
2020 – 2022	Nur Shahir	Postdoc	PSU	Biology
2016 – 2017	Xiaoling Gong	Scientist	Cornell	MBG

Undergraduate Student Research Mentoring

Years	Name	Institution	Program
2026 –	David Kunselman	PSU	Shreyer Honors College; Biology
2026 –	Ellie Marshall	PSU	Biology
2026 –	Stephen Evans	PSU	Biology
2025 –	Joshua Miller	PSU	Biochemistry
2025 –	Eliana Garza	PSU	Biochemistry
2025 –	Anwi Tadzong	PSU	Biomedical Engineering
2025 –	Katerina Novic	PSU	Biology
	---> *Awarded an Eberly College of Science Research Award (2026)*		
2024 –	Luscinia Quance	PSU	Schreyer Honors College; Biology
2026	Damien Perry	University of Houston	Summer Research Opportunities Program (SROP)
2026	Kinsley Wilson	Spellman	Summer Research Opportunities Program (SROP)
2025	Chris Kuetche	PSU	Biology
2023 – 2026	Jack Stevenson	PSU	Biology
2023 – 2026	Patricia Brocious	PSU	Schreyer Honors College; Biology
	---> *Awarded an Eberly College of Science Research Award (2025, 2026)*		
	---> *Awarded an Undergraduate Research Conference Presentation Travel Support award (2025)*		
	---> *Awarded a Schreyer Honors College Travel Award (2025)*		
	---> *Honors thesis title: "Using co-culture systems to disentangle direct vs. indirect effects of the microbiome on host gene expression regulation"*		
2022 – 2025	Maria Lovallo	PSU	Microbiology

---> *NASA Pennsylvania Space Grant Consortium Awardee (2022)*

---> *Profiled by the Huck Institute of the Life Sciences in 2025: https://www.huck.psu.edu/news/graduating-senior-reflects-on-undergraduate-research-experiences-at-penn-state*

2023	Muhammad Rujhan Najib Bin Fauzi Najib	PSU	Computer Science
2023	Evan Lloyd	PSU	Biology
2021 – 2023	Meera Gupta	PSU	Accelerated Premedical-Medical Program

---> *Awarded an Eberly College of Science Research Award (2022)*

2023	Steven Kadavil	PSU	Computer Science
2022 – 2023	Shivani Kumar	PSU	Biology
2021 – 2023	Aureo Zanon	PSU	Biology and Statistics, minor in Math

---> *Awarded an Erickson Discovery Grant (2022)*

2022	Mansi Chandra	Juniata	Bioinformatics
2022	Shanisha McGuire	PSU	Biochemistry and Molecular Biology
2021	Kyrah Williams	PSU	Biology, Millennium Scholar
2021 – 2022	Jovial Joseph	PSU	Accelerated Premedical-Medical Program

---> *Awarded an Eberly College of Science Research Award (2021)*

2021	Yiyan Zhang	PSU	Statistics – Biostats, Math – Systems Analysis
2021	Sean O'Rourke	PSU	Biology – Neuroscience
2018	Adon Chowdhury	Cornell	Biometry and Statistics
2017 – 2018	Trang Dau	Cornell	Human Biology, Health, & Society
2015 – 2017	Monica Guardado	PSU	Biology

---> *Awarded an American Society of Microbiology (ASM) Research Capstone Fellowship (2017)*

Rotation Advisor

Year	Program	Name(s)
2026	Bioinformatics and Genomics	Sanyam Shah
2023	Molecular, Cellular, & Integrative Biosciences	Hanh Tran, Jiwoo Kim
	Biology	Stephanie Won
2022	Biology	Naomi Huntley, Nicholas Locatelli
2021	Bioinformatics and Genomics	Kobie Kirven, Polina Tikhonova
	Biology	Annamaria Calderon
	Molecular, Cellular, & Integrative Biosciences	Cassandra Whitmoyer
	Biochemistry, Microbiology, and Molec. Biology	Anh Pham
2020	Bioinformatics and Genomics	Kyle McGovern
2019	Biology	Erica Ryu

Dissertation Committee Member

Defended	Deg.	Name	University	Department/Program
In progress	Ph.D.	Jasper Chao	PSU	Biology
In progress	Ph.D.	Jessica Buchser	PSU	Biology
In progress	Ph.D.	Jennifer Lausch	Penn State COM	Biomedical Sciences
In progress	Ph.D.	Udeshi Wickramarachchi	PSU	MCIBS
In progress	Ph.D.	Daisy Noe	PSU	Biology
In progress	Ph.D.	Dina Memar Moshrefi	PSU	Biology
In progress	Ph.D.	Yao Tu	PSU	MCIBS
In progress	Ph.D.	Tayyaba Alvi	The Leibniz Institute on Aging – (FLI)	Leibniz Graduate School on Aging
2026	Ph.D.	Jennifer Harris	PSU	Ecology
2026	Ph.D.	Anastasia Lucas	University of Pennsylvania	Genomics & Computational Biology
2026	Ph.D.	Shreya Ramachandran	University of Chicago	Human Genetics
2025	Ph.D.	Susan Tian	PSU	BMMB
2025	M.S	Shriya Gangakhedkar	Penn State COM	Biomedical Sciences
2024	Ph.D.	Sterling Wright	PSU	Anthropology
2024	Ph.D.	Lan-Nhi Phung	PSU	Biology
2024	Ph.D.	Scott Eckert	Penn State COM	Bioinfo. and Genomics
2023	M.S.	Julia Stewart	PSU	Biology
2023	Ph.D.	Jeremy Sutherland	PSU	Bioinfo. and Genomics
2022	M.S.	Miranda DePriest	PSU	Plant Path. and Env. Micro.
2022	MPS	Veronica Cappas	PSU	Forensic Science
2022	Ph.D.	Lixiang Zhang	PSU	Statistics
N/A	Ph.D.	Katrin Peterson	PSU	BMMB
N/A	M.S.	Nellie Heitzman	Bucknell University	Biology

Qualifying Exam Committee Member

Year	Program	Students
2025	Bioinformatics and Genomics	Nikol Chantzi, Tram Ha
2023	Bioinformatics and Genomics	Salaar Khan
2022	Bioinformatics and Genomics	Avantika Diwadkar
2021	Bioinformatics and Genomics	Corrine Smolen
	MCIBS	Sofia Kenney
2020	Bioinformatics and Genomics	Jeremy Sutherland, Shaopeng Liu

Undergraduate Thesis Reader

Year	Program	Students
2026	Science Research Distinction Certificate	Caroline Nowicki

Other

Year	Position	Person
2023	Visiting Professor from Colgate University	Bineyam Taye

Academic Honors

2025	Pennsylvania State University One Health Microbiome Center Rising Star Research Award
2024	Applied Microbiology International WH Pierce Prize for Global Impact in Microbiology (to Pennsylvania State University One Health Microbiome Center)
2007	Graduated with comprehensive honors: honors in Bacteriology and the liberal arts (University of Wisconsin – Madison)
2004	Dean's List (University of Wisconsin – Madison)
2003	William F. Vilas Scholarship (University of Wisconsin – Madison)

Professional Development

2025	**Pennsylvania State University T32 Mentor-Mentee Summit** – Mentor training
2024	**Pennsylvania State University T32 Mentor-Mentee Summit** – Mentor training
2024	**COMPASS Scholars Program** – NIH R25GM143346 funded program based at Washington University in St. Louis that offers national training and mentoring program for early career faculty
2021	**Safer People Safer Places Transgender and Gender Inclusion 101** – Pennsylvania State University
2020	**Becoming an Active Witness who Hears, Sees, Feels, and Does** active bystander training – Pennsylvania State University
2019	**Mentoring Matters Workshop** – Eberly College of Science, Pennsylvania State University
2017 – 2018	**Cornell Center for Teaching Excellence GET SET teaching and learning workshops completed**: - An Integrated Course Design Approach to Planning Your Class

- Tips on Writing a Strong Teaching Philosophy Statement
- Integrating Technology into Your Classroom
- Developing Service-Learning in the Disciplines
- Flipping the Classroom as a TA
- Using Your Research Experience to Improve Your Teaching
- Holding Effective Office Hours
- Teaching and Mentoring Across Differences
- Leading a Discussion in an Online Classroom

2017 **Assessing Learning and Teaching certificate** – Cornell University Center for Teaching Innovation. Workshops included:
- Designing Learning Outcomes for Your Course
- Utilizing Classroom Assessment Techniques to Evaluate Student Learning
- Developing Rubrics for Effective Grading

2017 **Creating an Engaging Classroom certificate** – Cornell University Center for Teaching Innovation
- Building a Collaborative Learning Environment
- Using Theater Techniques to Enhance Your Teaching
- Engaging Students in Quantitative Courses

2016 **The Practice of Inclusive Teaching in STEM certificate** – Cornell University Center for Teaching Excellence

2016 **Building Mentoring Skills for an Academic Career certificate program** – Cornell University Center for the Integration of Teaching and Learning (CU-CIRTL)

2015 - 2016 **Postdoc Leadership Certificate Program** – Cornell University

2014 **Certificate in University Teaching** – University of Chicago Center for Teaching and Learning

2013 **Software Carpentry Instructor training** – Software Carpentry

Professional Affiliations

2026 Society for Integrative & Comparative Biology (SICB)

2017 – Genetics Society of America (GSA)

2014 – Society for Molecular Biology and Evolution (SMBE)

2012 – American Society of Human Genetics (ASHG)

2017, 2022 – American Society for Microbiology (ASM)

2019 – 2022 Society for Advancement of Chicanos/Hispanics and Native Americans in Science (SACNAS)

2017, 2026 American Association of Physical Anthropologists (AAPA)

2016 – 2019 National Postdoc Association (NPA)

2011 – 2018 American Association for the Advancement of Science (AAAS)

Outreach

2026 **Panelist** – "From Protocols to Pipelines: Connecting Bioinformatics with Experimental Biology" 10th Annual Life Sciences Symposium, Huck Institutes of the Life Sciences, Pennsylvania State University

2025 **Speaker** – "Causes and consequences of human gut microbiome composition"; SSP International Genomics Summer Science Program; Lehigh University

2023 **Speaker** – "Unlocking the Hidden Universe Within: Exploring the Astonishing World of the Human Microbiome"; Foxdale Village Retirement Community

2022 **Speaker** – Investigating Careers and Networking (iCAN) Speaker Series "Establishing your Academic Research Career"; Johns Hopkins University

2022 **Interview** – Pennsylvania State University's Center for Human Evolution and Diversity Tracking traits podcast: *A Geneticist's Approach to Human Microbiomes:* https://trackingtraits.buzzsprout.com/1762618/11778386-a-geneticist-s-approach-to-human-microbiomes?t=0

2021 **Panelist** – Microbiome Center Consortium Seminar Series NMDC Ambassadors Forum

2021 **Advocate** – American Society of Human Genetics Hill Day advocate

2020 **Interview** – Eberly College of Science "Hey! I Got a Question About That" podcast on the microbiome: https://www.youtube.com/watch?v=960zTLUD7VU&t=881s

2019 **Advocate** – Rally for Medical Research Hill Day advocate, representing American Society of Human Genetics

2015 – 2022 **Reviewer** – American Society of Human Genetics (ASHG) DNA Day Essay Judge

2015 – 2018 **Member** – Genetics Education and Outreach Network (GEON)

Leadership and Service

Departmental

2020 – **Member** – Seminar Series Organizing Committee, Department of Biology, Pennsylvania State University

2020 – 2023, 2025 - **Member** – Graduate Admissions Committee, Department of Biology, Pennsylvania State University

2023 – 2025 **Member** – Biology Climate and Diversity Committee, Department of Biology, Pennsylvania State University

2010 - 2013 **Representative** – Molecular Biosciences organizational committee: student Representative from the Department of Human Genetics on orientation week,

annual molecular biosciences retreat, and recruitment organizing committees, Department of Human Genetics, University of Chicago

Institutional

2026 **Organizer and participant** – NIH R35 renewal/conversion workshop series, Pennsylvania State University

2026 **Discussion leader** – Eukaryotic Gene Regulation (EGR) T32 Training Grant Ethics Discussion, Pennsylvania State University

2026 **Member** – AC-14 Review Committee (Department Head 5-year review), Eberly College of Science, Pennsylvania State University

2025 **Instructor** – Biotechnological & Integrative Opportunities in Microbiome Sciences (BIOMES) T32 Training Grant Ethics Workshop, Pennsylvania State University

2023 – **Member** – Molecular, Cellular, and Integrative Biosciences (MCIBS) Graduate Program Qualification Exam Committee, Pennsylvania State University

2024, 2026 **Poster Session Judge** – One Health Microbiome Center Symposium, Pennsylvania State University

2025 **Member** – Eukaryotic Gene Regulation (EGR) training grant selection committee, Pennsylvania State University

2021 – 2022 **Member** – Institute for Computational and Data Sciences (ICDS) Coordinating Committee, Pennsylvania State University

2021 – 2022 **Co-chair** – "Changing Microbiomes" Conference Organizing Committee, Pennsylvania State University

2021 **Member** – Microbiome Center Director Search Advisory Committee, Huck Institutes of the Life Sciences, Pennsylvania State University

2021 **Invited Panelist** – "First Years as Faculty in STEM" GRADUCon, University of Chicago

2020 – 2021 **Grant reviewer** – Pennsylvania State University and University at Buffalo Clinical and Translational Science Pilot Award Program, Pennsylvania State University

2018 **Member** – Consensual Relationships Policy Committee – Postdoc representative, Cornell University. See http://theuniversityfaculty.cornell.edu/news/consensual-relationships-policy-committee/ for details.

2017 – 2018 **Cofounder & Organizer** – Postdoc Fellow Invited Lecture in Research and Career Development series. Department of Molecular Biology, Cornell University

2016 – 2018 **Member** –Postdoctoral Advisory Council, Cornell University

2016 – 2017 **Reviewer** – Sigma Xi Grants-In-Aid of Research, Cornell University

National/International

Leadership and Service to Scientific Societies

2023 – 2025 **Member** – National Microbiome Data Collaborative (NMDC) Ambassador Selection

Committee

2022 – 2025 **Member** – Genetics Society of America DeLill Nasser Award Selection Committee

2018 – 2025 **Member** – Federation of American Societies for Experimental Biology (FASEB) Training & Career Opportunities Subcommittee

2015 – 2023 **Judge** – ASHG DNA Day Essay Contest

2021 – 2022 **Ambassador** – National Microbiome Data Collaborative (NMDC)

2020 – 2021 **Co-chair** – American Society of Human Genetics (ASHG) Career Development Committee (CDC)

2019 **Incoming Chair** – American Society of Human Genetics (ASHG) Training and Development Committee

2017 – 2019 **Member** – American Society of Human Genetics (ASHG) Training and Development (TDC) Committee

Grant Review

2023, 2026 ***Ad hoc* Member** – NIH MRAF study section

2020 – 2022, 2024, 2025 **Grant Reviewer** – National Science Foundation (NSF)

2022 **Reviewer** – Swiss National Science Foundation

2021 ***Ad hoc* Member** – NIH GCAT study section, Early Career Reviewer Program

2018, 2020 **Reviewer** – Wellcome Trust Investigator Award in Science

Conference Related Service

2019 – 2021 **Abstract Reviewer** – SACNAS research presentation abstracts (Life Sciences)

2019 – 2021 **Abstract Reviewer** – SACNAS Travel Scholarships (Life Sciences)

2020 **Moderator** –The Allied Genetics Conference (TAGC) platform session: Disease Models and Aging (Mammal) *TAGC 2020 was held virtually due to COVID-19

2019 **Moderator** – American Society of Human Genetics (ASHG) Annual Meeting platform session: Gene Expression Variation Across Diverse Global Populations

2019 **Abstract Reviewer** – American Society of Human Genetics (ASHG) Annual Meeting abstracts (Bioinformatics and Computational Approaches)

2018 **Session Chair** – Probabilistic Modeling in Genomics (ProbGen) Meeting at Cold Spring Harbor Labs

2017 **Moderator** – Academic Career Panel at the American Society of Human Genetics Annual Meeting (ASHG)

2011, 2012, 2014 **Judge** – Annual Chicago Public Schools Student Science fair

2011 - 2014 **Judge** – Annual Chicago Area Undergraduate Research Symposium (CAURS)

Manuscript Reviewer

2015 –

- Annals of Applied Statistics
- American Journal of Primatology
- Applied and Environmental Microbiology (AEM)
- Biology Letters
- BMC Genomics
- Cell
- Cell Host & Microbe
- Cell Multi-Journal Submission
- Cell Reports
- Communications Biology
- Critical Reviews in Microbiology
- Diabetologia
- eLife
- Environmental Microbiology
- Genes
- Genome Biology
- Genome Research
- Genomics, Proteomics, and Bioinformatics
- Gut Microbes
- ISME Host Microbe
- Journal of Allergy and Clinical Immunology (JACI)
- Journal of Evolutionary Biology
- JoVE
- Microbial Ecology
- Microbial Pathogenesis
- Microbiome
- Microorganisms
- mSystems
- Nature Communications
- Nature Microbiology
- npj Biofilms and Microbiomes
- PLoS Genetics
- PLoS ONE
- PNAS
- Science
- Scientific Reports
- Thorax
- Trends in Genetics
- Trends in Immunology

Other

2025 – **Letter Writer for Promotion and Tenure** – ETH Zurich

2019 – **Board Member** – Cards Against Humanity Science Ambassador Scholarship Program